

Matthew S. Euliano

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SUMMARY

Senior software engineer and technical lead specializing in robotics, simulation, computer vision, real-time telemetry, and hardware/software integration. Built digital-twin, AR, AI/CV, and replay tools for surgical robotics R&D, turning ambiguous research needs into reliable prototypes, study tools, and executive demos.

TECHNICAL SKILLS

Languages / Systems: C#, Python, C++, MATLAB, APIs, Windows/Linux, Git/CMake, DDS/RTI

Robotics / CV: stereo calibration, kinematics, ROS, OpenCV, sensor fusion, SLAM, segmentation/redaction

Simulation / 3D: Unity, Unreal Engine 4, Gazebo, Isaac Sim, OpenUSD, NVIDIA Omniverse, ShaderLab/HLSL

Data / AI: ONNX, telemetry replay, real-time visualization, GPU optimization, multi-sensor workflows

EXPERIENCE

Medtronic | Senior Software Engineer — Boston, MA Jan 2023 – Jan 2026

- Led robotics visualization, simulation, digital-twin, and AR initiatives for surgical robotics R&D, turning research and user needs into requirements, prototypes, study tools, and executive demos.
- Built a high-throughput DDS logging/playback and digital-twin replay platform for robotic telemetry, enabling deterministic case review across 8+ hospitals and 25+ labs.
- Built real-time multi-sensor visualization workflows for stereo calibration, distortion correction, video alignment, multi-display rendering, and hardware/software interfaces.
- Integrated ONNX AI inference and native C++/CUDA simulation into prototypes for segmentation, redaction, and real-time deformation visualization.
- Presented technical strategy and prototypes to surgeons, executives, conferences, and industry audiences; contributed to 10+ patent disclosures.

TruPhysics GmbH | Robotics Software Engineer — Stuttgart, Germany Jun 2022 – Aug 2022

- Developed ROS packages and Python wrappers for a pick-and-place 6-DOF robotic arm and industrial mobile robot in Linux.
- Integrated a multimodal sensor suite on a linear axis to assess material quality; built a monitoring UI for real-time performance and data capture.

Northeastern University RIVeR Lab | Graduate Research Assistant — Boston, MA Feb 2022 – Dec 2022

- Implemented mapping, localization, and navigation behaviors for a mobile manipulation platform using ROS Navigation Stack with the Timed-Elastic Band (TEB) local planner.
- Programmed human-following behavior in simulation using Python and a leg-detection pipeline with RGB-D and laser sensing.

L3Harris Unmanned Systems | Systems Integration Engineer — Ashburn, VA Jul 2019 – Dec 2021

- Tested and integrated small jet engine systems across ground and flight test, controls verification, and performance analysis; executed HIL/SIL and final acceptance testing.

EDUCATION

Northeastern University — M.S. Robotics — Boston, MA (GPA 3.97) Dec 2023

Virginia Tech — B.S. Mechanical Engineering — Blacksburg, VA (GPA 3.63) May 2019